Programming(1)

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Learning Objectives

- ✓ Identify and understand different data types
- ✓ Use boolean values (True / False)
- ✓ Format text with f-strings
- ✓ Create and modify lists



Boolian



Boolian

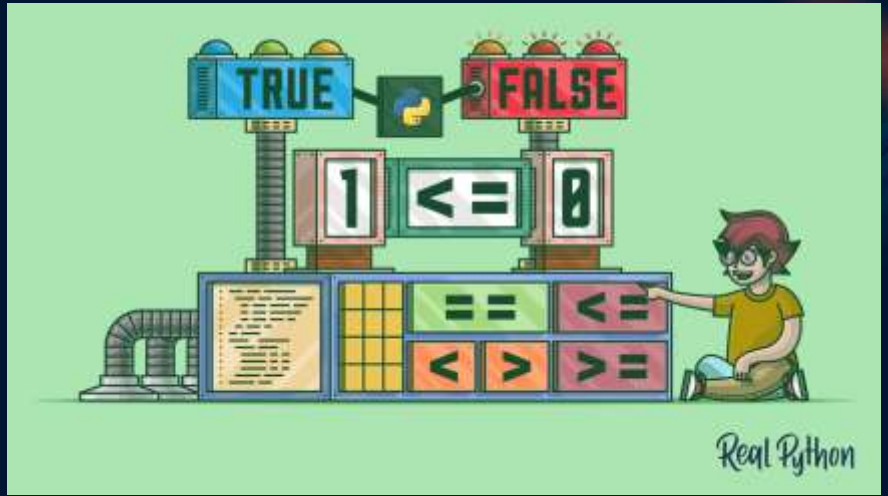
Booleans are used for logical decisions

index.py > ...

```python
is_sunny = True
is_raining = False
print(is_sunny)

x = 5
print(x > 3)    # True
print(x == 10) # False

```



f-string



f_string

index.py > ...

```python
name = "Ahmad"
age = 20
print(f"My name is {name} and I am {age} years old.")


print("My name is", name, "and I am", age, "years old.")

```



Datatypes



List

Used to store multiple values in one variable

Written with square brackets []



List Example

index.py > ...

```python
fruits = ["apple", "banana", "cherry"]
print(fruits)

print(fruits[0])   # apple
print(fruits[-1])  # cherry

```



Modifying Lists

Lists are mutable → you can change them.

index.py > ...

```python
fruits = ["apple", "banana", "cherry"]


fruits.append("orange")   # add item
fruits.remove("banana")   # remove item
print(fruits)

```



Tuple



PYnative.com

Tuples in Python

T = (20, 'Jessa', 35.75, [30, 60, 90])

T[0] T[1] T[2] T[3]

- ✓ **Ordered**: Maintain the order of the data insertion.
- ✓ **Unchangeable**: Tuples are immutable and we can't modify items.
- ✓ **Heterogeneous**: Tuples can contains data of types
- ✓ **Contains duplicate**: Allows duplicates data

Set



PYnative.com

Set in Python

S = { 20, 'Jessa', 35.75}

- ✓ **Unordered**: Set doesn't maintain the order of the data insertion.
- ✓ **Unchangeable**: Set are immutable and we can't modify items.
- ✓ **Heterogeneous**: Set can contains data of all types
- ✓ **Unique**: Set doesn't allows duplicates items

Dict



Dictionary in Python

PYnative.com

Unordered collections of unique values stored in (Key-Value) pairs.

```
d = {'a': 10, 'b': 20, 'c': 30}
```

d['a'] d['b'] d['c']

- ✓ **Unordered**: The items in dict are stored without any index value
- ✓ **Unique**: Keys in dictionaries should be Unique
- ✓ **Mutable**: We can add/Modify/Remove key-value after the creation

Thank You

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